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The Intensity of the Tertiary Geomagnetic Field

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Summary

Although many paleomagnetic directional data are now available, very little is known of the ancient intensity of the Earth's magnetic field. This paper describes the determination of the Tertiary geomagnetic field intensity using baked contacts (red sandstones, conglomerates and dykes) and dykes from Scotland and lavas and baked laterites from Iceland.

The field intensity results obtained correspond to a mean Tertiary dipole moment of $4{\cdot}94 \times 10^{25}$ gauss.cm^3 in Scotland and a mean Tertiary dipole moment of $5{\cdot}34 \times 10^{25}$ gauss.cm^3 in Iceland. These values contrast with the present geomagnetic dipole moment which is $8{\cdot}0 \times 10^{25}$ gauss.cm^3.

1. Introduction

The direction of the Earth's main magnetic field at various localities and over the past several hundred million years has been deduced from paleomagnetic measurements on rocks of many types. Although a fairly complete record now exists of field direction since the Pre-cambrian [Irving 1964, Irving (& Stott) 1960–1965], comparatively little is known of the intensity of either the dipole or non-dipole geomagnetic fields over the period covered by the rock record. This is certainly because the experimental difficulties encountered in the determination of field intensity from rocks and the uncertainties in the interpretation of the results obtained are much greater than in the corresponding measurements of direction only.

Nevertheless some field intensity results have been published, and intensity values may be derived from data presented by other authors. For the geological time scale, field intensity determinations have been presented by Momose (1963) (Upper Miocene–Upper Pliocene), Van Zijl (1961) (Triassic–Jurassic) and Briden (1966) (late Silurian–early Devonian). Approximate intensity values may also be deduced from data presented by Johnson *et al.* (1948) (Quaternary), Nagata (1943) (Pleistocene), Koenigsberger (1938) (Tertiary and Permo–Carboniferous) and Clegg *et al.* (1954) (Triassic). In addition, intensity determinations on single, or pairs of, geological specimens have been carried out by Thellier *et al.* (1959) (Quaternary), Weaving (1960) (Tertiary) and Wilson (1961) (Tertiary).

In this paper the results of field intensity determinations from Scottish Tertiary baked contacts and dykes and Icelandic Tertiary baked laterites and lavas will be described.

2. Basis of field intensity determination

It was shown by Nagata (1943) for thermoremanent magnetization (TRM) and by Johnson *et al.* (1948) for detrital remanent magnetization (DRM) that for fields

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of about one oersted or less the intensity of magnetization acquired by a rock is proportional to the field in which it is laid down. Thus if M_n is the natural remanence (NRM) acquired by either TRM or DRM in an ancient field F, and M_a is an artificial remanence induced in the same specimen by the same magnetization process in a field H, it follows that:

$$\frac{F}{H} = \frac{M_n}{M_a}. \tag{1}$$

This is the basic equation for field intensity determination. Results published hitherto (and those to be presented here) have all indicated the Earth's field to have been less than one oersted, and so the validity of this formula is not in doubt for the case of the Earth. In practice, however, it is seldom possible to apply equation (1) directly for two reasons. Firstly, the NRM or the rock may have changed since its acquisition so that the measured value of M_n may be considerably different from the original NRM intensity induced in field F. It is possible that partial TRMs, caused by slight heating (for example, by thermal waters) and cooling in the Earth's field, may have been superimposed, or that the original NRM may have decayed to some extent followed by the acquisition of a viscous remanent magnetization (VRM) in the Earth's field (Rimbert 1958). Secondly, during the process of inducing the artificial remanence the nature and amount of magnetic material present may be altered so that the initial and artificial magnetizations are imparted to essentially different substances. This is particularly likely to arise during the heating and cooling necessary to produce an artificial TRM, when oxidation or reduction may occur (Akimoto *et al.* 1957, Akimoto *et al.* 1959, Havard *et al.* 1965, Ade-Hall *et al.* 1965).

In the present work all magnetizations were TRMs, and hence the main technical problems were to determine the extent of change of the NRM and to determine the degree of alteration occurring during the laboratory heating. These problems will be discussed in greater detail below.

3. Treatment of field intensity data

It is clear from spherical harmonic analysis of the geomagnetic field (Gauss 1838, and many others since) that the dipole term is predominant; and hence to a first approximation the basic formula relating the magnetic inclination (I) of the Earth's field to latitude (λ) is $\tan I = 2 \tan \lambda$. In palaeomagnetism I is usually measured and the ancient site latitude is calculated from this formula. Strictly speaking the latitude calculated is the magnetic latitude. In practice, however, it is commonly assumed that averaged over long periods ($>10^4$ years) the Earth's dipole has been axial so that the calculated latitude is taken as the geographic latitude (Runcorn 1959, Creer 1962, Irving 1964, and others) and the effects of non-dipole fields or wobble of the main dipole are treated as 'noise' superimposed on the main dipole 'signal' (Cox *et al.* 1960).

For a dipole of moment P the field intensity H produced at radial distance r and latitude λ is given by:

$$H = Pr^{-3}(4 - 3\cos^2\lambda)^{\frac{1}{2}}. \tag{2}$$

This equation indicates that if the geomagnetic field has always been essentially dipolar the ancient field intensity deduced from rock data will be a function of latitude. It would be convenient, therefore, to have some method of treating intensity results which would enable values from sampling sites at different latitudes to be compared directly, with the minimum of assumptions regarding the past behaviour of the field. It is proposed here to use the intensity values obtained to calculate the equivalent dipole moment which would have produced the measured intensity at the palaeolatitude

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of the sample (equation 2). This is analogous to the calculation of virtual geomagnetic pole positions (VGP) from paleodirectional data; and hence the calculated dipole moment will be called the *virtual dipole moment* (VDM).

One advantage of this method is that no scatter is introduced by any wobble of the main dipole since the determined magnetic palaeolatitude is independent of the orientation of the dipole relative to the Earth's rotational axis. It is thus not necessary to make use of the axial field assumption. Secular field variations will introduce scatter and hence as in directional studies it is necessary to utilise a large number of specimens covering a period of at least 10^4 years to mean out the effects of the non-dipole field.

4. Ancient field intensities: Scottish Tertiary suite

4.1. *Rocks used and their suitability*

Ancient field intensity determinations have been carried out on twenty-seven specimens from seventeen contacts baked by dykes during the Tertiary in Scotland. These contacts comprised mainly red sandstones and conglomerates, but the collection also included basalt dykes and grey fine-grained sediments. In all cases the contacts were taken from within six cm of the dyke edges in order to ensure that only the most highly baked material was utilized. Collection localities, sampling details, directions of magnetization of the contacts and of the dykes which baked them, and potassium–argon ages of certain of these dykes have previously been reported (Smith 1966a). The eight dated dykes possessed ages lying in the range 33·9 m.y. to 57·1 m.y.

Preliminary considerations indicated that baked rocks are among the most suitable for field intensity work. Rocks with low Curie points tend to oxidize on heating (Ade-Hall *et al.* 1965) with a consequent undesirable increase in Curie point, so that an initially high Curie point is usually a necessary property. All the baked contacts in this study possessed high field Curie points of 500 °C or over and fifteen of the seventeen Curie points were 570 °C or over. Since field baking had already occurred it was likely that much of the alteration likely to occur during heating would have taken place before the acquisition of the TRM. Further, since the original chemical remanent magnetization (CRM) or DRM had been replaced by TRM the intensity of magnetization was about 100 times larger than that of the adjacent unbaked sediment and hence measurable to a higher degree of accuracy. The baked contacts were also highly stable against viscous decay and remagnetization (Smith 1966a). The mean change in direction of magnetization during thermal demagnetization for the particular specimens used for intensity determination was only 5°. Most secondary components had disappeared below 100 °C, as evidence by constancy of direction above this temperature in the vector heating diagrams.

4.2. *Basic intensity determination*

Each specimen was heated in zero field to above its Curie point and then cooled in a field of 0·60 oersted. Measurements of the NRM and TRM were made at 50 °C intervals between 20 °C and 670 °C using an astatic magnetometer. It was then a simple matter to apply equation (1) at any temperature T, where M_n and M_a then referred to the partial NRM and TRM, respectively, acquired between temperature T and the Curie point. The three temperatures 270 °C, 320 °C and 370 °C were chosen since they were well above the effects of viscous components. At temperatures higher than 370 °C the moments became too small and changed too rapidly with temperature to enable them to be measured accurately.

All specimens were heated and cooled in air. It might be thought that heating and cooling in a vacuum would be a guarantee against chemical change, but this is not necessarily so. Even in a vacuum oxidation or reduction may occur in the presence of other minerals (Havard 1965). In view of the impossibility of guarding against

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Table 1

Basic intensity data for the Scottish baked contacts and dykes. F=ancient field intensity; σ=estimated standard deviation of F values at three temperatures.

Type	No.	Specimen	F (oe) (370°C)	F (oe) (320°C)	F (oe) (270°C)	mean F (oe)	σ (oe)	σ %
Baked contact	1	IVA1	0·19	0·19	0·19	0·19	0·000	0·0
	2	ID1	0·27	0·26	0·23	0·25	0·021	8·4
		ID2	0·31	0·30	0·27	0·30	0·022	7·3
	3	IVD1	0·17	0·18	0·17	0·17	0·007	4·1
	4	IVE2	0·30	0·35	0·40	0·35	0·050	14·3
	5	VF1	0·25	0·26	0·27	0·26	0·010	3·8
		VF2	0·26	0·30	0·29	0·28	0·021	7·5
	6	IVG2	0·56	0·59	0·57	0·57	0·016	2·8
		VG2	0·44	0·48	0·44	0·45	0·023	5·1
	7	IVK2	0·24	0·24	0·26	0·25	0·010	4·0
	8	IIIL1	0·20	0·20	0·20	0·20	0·000	0·0
	9	IIIM2	0·24	0·25	0·32	0·27	0·044	16·3
	10	IIN1	0·23	0·22	0·20	0·22	0·016	7·3
		IIN2	0·28	0·26	0·27	0·27	0·010	3·7
	11	IIP2	0·20	0·22	0·25	0·22	0·026	11·8
		IVP2	0·22	0·27	0·31	0·27	0·045	16·7
	12	IQ1	0·51	0·64	0·69	0·61	0·093	15·2
		IQ2	0·57	0·66	0·79	0·67	0·112	16·7
	13	IVS2	0·47	0·54	0·57	0·53	0·052	9·8
		VS1	0·47	0·49	0·54	0·50	0·036	7·2
	14	IIIU2	0·17	0·20	0·21	0·19	0·021	11·1
		IVU2	0·16	0·18	0·20	0·18	0·020	11·1
	15	IVV1	0·47	0·45	0·53	0·48	0·042	8·7
		IVV2	0·35	0·43	0·48	0·42	0·066	15·7
	16	IVY1	0·10	0·083	0·071	0·085	0·015	17·6
		IVY2	0·083	0·073	0·064	0·073	0·010	13·7
	17	IVZ1	0·058	0·062	0·049	0·056	0·006	10·7
Dyke	14a	IU2	0·16	0·22	0·29	0·22	0·065	29·5
	18	IIW1	0·17	0·19	0·22	0·19	0·025	13·1

alteration in rocks where alteration was likely to occur, it was necessary to use rocks which were highly stable against chemical change. In practice, this meant using naturally highly oxidized rocks and heating them in an oxidizing atmosphere (air). The problem of alteration will be discussed further in Section 4.4.

The field intensity value for each of the twenty-seven specimens at each temperature is given in Table 1, which also shows the mean of the intensities at the three temperatures and the estimated standard deviation over the three temperatures. (In this paper standard deviations have been calculated using the normal formula with Bessel's correction. It is realised that for a small number of values the calculated standard deviations may not be accurate but will be the best estimates possible.)

4.3. *The problem of secondary components*

It frequently occurs that the NRM a rock possesses at the time of measurement is not the same as the moment acquired in the ambient field during its formation. Decay or removal of part of the original moment may have taken place followed by the superposition of one or more secondary components. These secondary components could be partial TRMs acquired during subsequent heating or viscous magnetizations produced in the Earth's field over long periods of time (Rimbert 1958). It is therefore necessary to consider the validity of the measured quantity M_n in equation (1).

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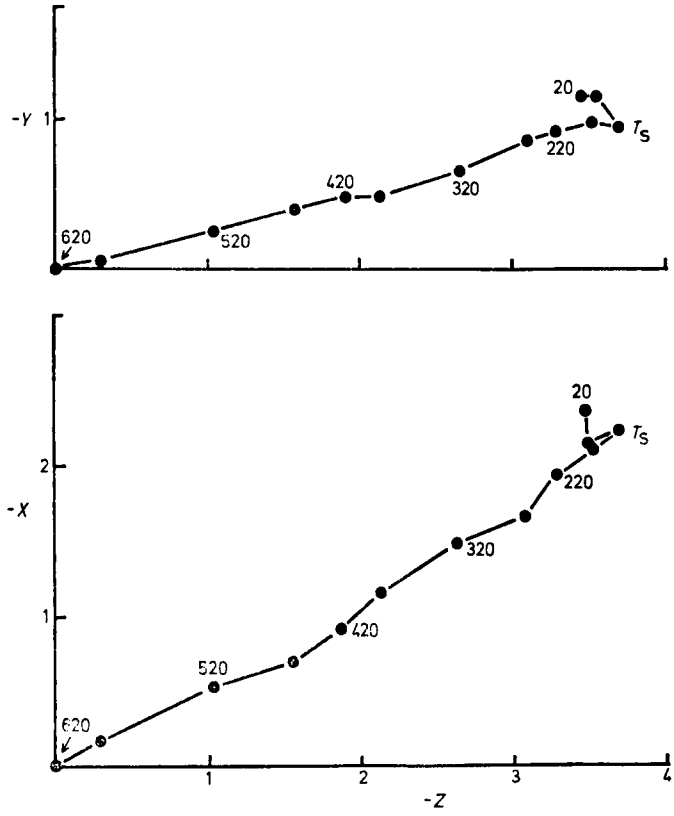


FIG. 1. Vector heating diagrams for Scottish baked sandstone (IIP2) in terms of three orthogonal components of magnetization (X, Y, Z in arbitrary units). Points are at 50 °C intervals; and numbers adjacent to points represent temperatures in °C. T_s is the approximate temperature of erasure of secondary components.

It is important to be clear on what effect the decay has on the original NRM. Fig. 1 shows the vector heating diagrams from one of the Scottish baked contact specimens. It is evident from these diagrams that the original moment has partly decayed and that a secondary component has been superimposed. In the present work the effects of viscous components were avoided by removing them by thermal demagnetization and then utilizing that part of the primary component remaining above the temperature of viscous erasure T_s (Fig. 1). But the question arises: has the moment above T_s decayed in any way, or in other words, is the moment remaining after the demagnetization to T_s the same as that possessed by the rock after reaching T_s during the original cooling?

Rocks usually possess a wide spectrum of blocking temperatures; and the lower blocking temperatures will be first to be affected by slight heating and the presence of weak fields for large periods of time in a direction different from that of the original magnetization. If T_b is the lowest blocking temperature not affected by such changes it is assumed here that $T_b \sim T_s$ and that above T_s no decay has occurred.

There is no evidence to suggest that decay has occurred above T_s, but there is some to suggest that decay is unlikely to have occurred. Rimbert (1958) and Van Zijl (1961) have shown that an isothermal remanent magnetization (IRM) artificially

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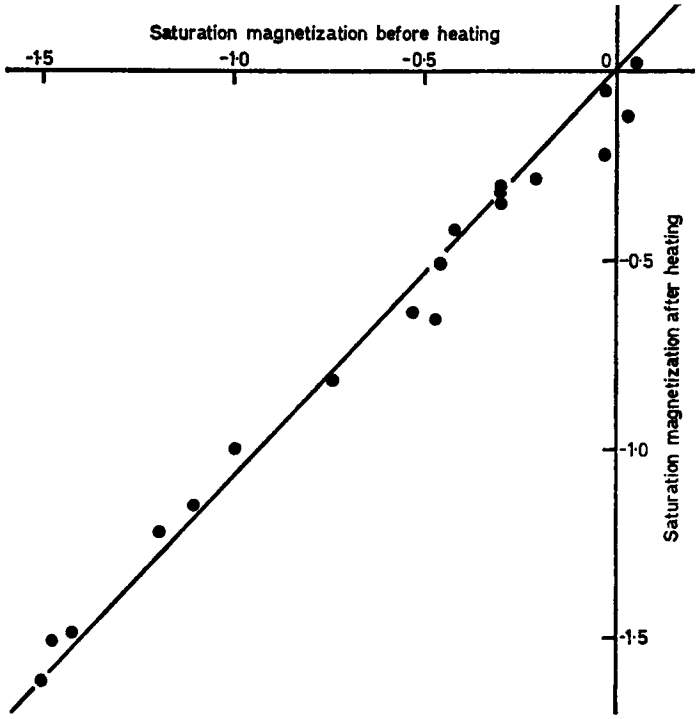


FIG. 2. Scottish baked contacts: saturation magnetization before heating plotted against that after heating to above the Curie point. J_{sat} in e.m.u./g plotted on $\log_{10}$ scale. Line represents mean change in J_{sat} of $-13{\cdot}0\%$.

induced in a rock may be completely removed by a.c. demagnetization without affecting the intensity of the NRM which remains above the highest a.c. field required. Also, it will be shown below that although the mean Tertiary field was smaller than the present field, the intensity values obtained from some individual specimens are higher than present values. Viscous decay cannot account for this.

It is thought, therefore, that as long as equation (1) is applied at temperatures above T_s the problem of secondary components can be overcome. This could be proved directly if many cases of agreement between field intensity values from baked contacts and the igneous bodies baking them could be obtained, since it would be expected that igneous and baked material would possess different decay characteristics in general. At present this evidence is not available.

4.4. *The problem of magnetic alteration during heating*

The method of intensity determination used here entailed heating directly to the Curie point followed by cooling in a 0·60 oersted field. It thus differed from the well-known technique developed by Thellier *et al.* (1959)* in which heating and cooling are carried out in temperature steps. By the present method it was not possible to calculate any field intensity value before the rock had been heated above its Curie point and hence not possible to use the consistency of intensity values from different temperature intervals to infer the absence of alteration.

* This is a review of work carried out by Thellier & Thellier up to 1959 and originally published in 1937a, b, 1938, 1941a, b, 1942 and 1946.

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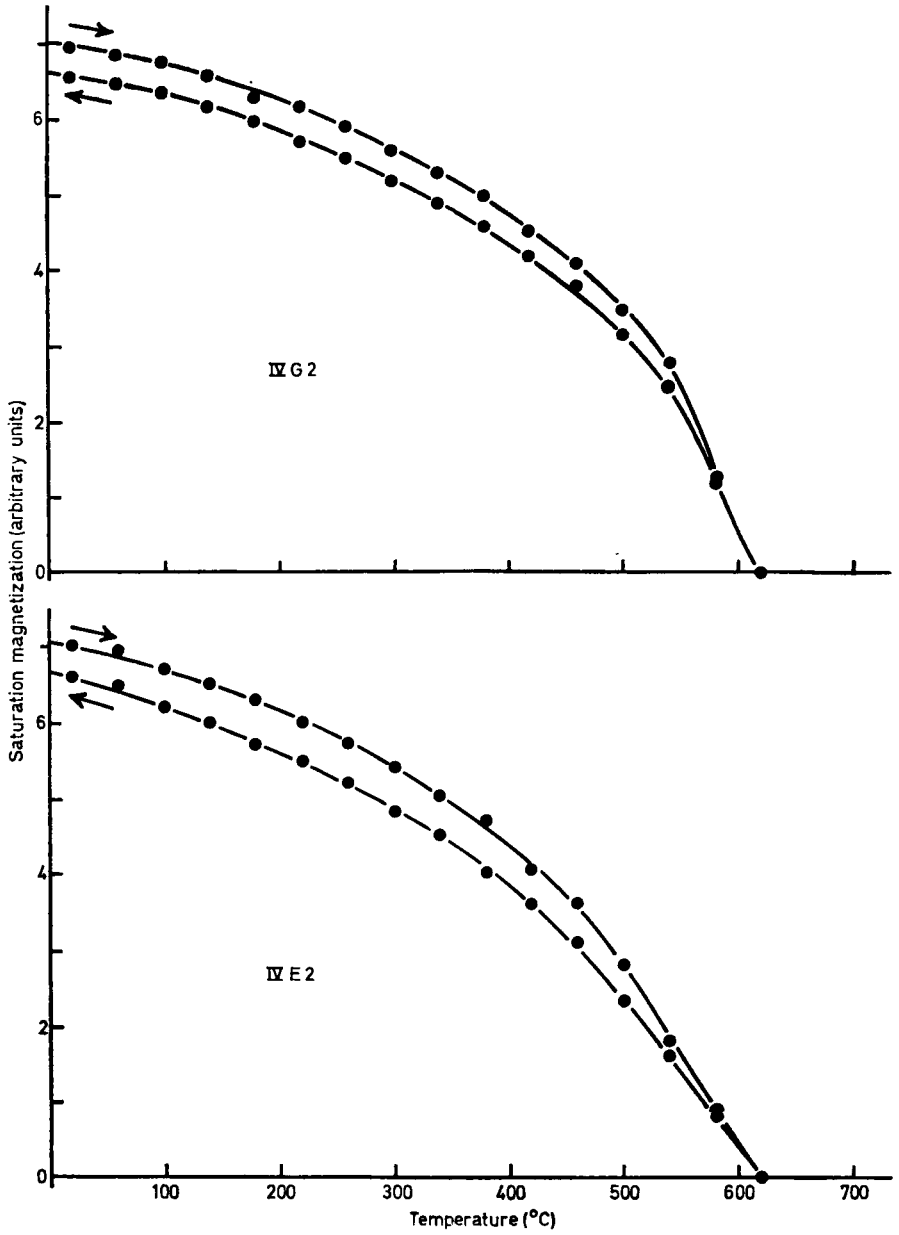


FIG. 3. Scottish baked contacts: examples of saturation magnetization heating and cooling curves.

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In the present work in order to estimate the effects of chemical or phase changes during heating the following were observed:

(a) high field (5000 oersteds) Curie points before and after heating,

(b) saturation magnetizations before and after heating,

(c) repeatability of the J_{sat} heating curves,

(d) Curie points before and after heating, measured in zero field,

(e) shapes of the NRM and TRM curves.

No one of these tests was alone sufficient to prove or disprove the presence of alteration. For example, although for a given specimen the saturation magnetization may not change on heating this is no guarantee of the validity of the intensity results because the NRM and the saturation magnetization may not have resided in the same magnetic constituent. It was thus necessary to apply as many tests as possible in order to estimate the degree of alteration.

(i) *High field characteristics*

The saturation magnetization for one specimen from each of the seventeen baked contacts was measured in a field of 5000 oersteds before and after heating in air. These values are plotted in Fig. 2. The mean change expressed as a percentage of the original saturation magnetization was $-13{\cdot}9\%$.

Wherever possible full J_{sat}–T curves were plotted, but in eight cases the specimens were too weakly magnetized to enable this to be done. In the other nine cases the Curie points did not change on heating within experimental error ($\pm 10\,^{\circ}C$), and the heating and cooling curves showed no dissimilarities in shape. Two examples of J_{sat}–T curves are shown in Fig. 3.

(ii) *Zero field characteristics*

For all twenty-seven specimens, zero field Curie points were deduced from the vector heating diagrams during heating and cooling of the natural moment. In no case was there a Curie point change within experimental error ($\pm 40\,^{\circ}C$ since zero field Curie points were often poorly defined) and in no case did the NRM and TRM curves differ in shape. Two examples of NRM–TRM curves are shown in Fig. 4.

(iii) *Dyke determinations*

The dykes baking the contacts were generally unsuitable for field intensity determination. They possessed Curie points in the range 350 °C–450 °C which rose to 550 °C–620 °C during heating in air. Nor was it generally possible to obtain field intensities by heating only to the Curie points of the dykes for the viscous components usually extended to about 270 °C which left too small a temperature interval between T_s and the Curie point over which to apply equation (1). However, the specimens from two dykes, W (for which the contacts were too weakly magnetized to measure) and U (for which contacts were measured) possessed viscous components which were removed by 100 °C. Field intensity values were obtained from one specimen from each of these dykes by heating only to their Curie points (375 °C for W and 450 °C for U) and applying equation (1) at 120 °C, 170 °C and 220 °C. These results are included in Table 1.

For the dyke-contact system U, we have:

dyke U: $F = (0{\cdot}22 \pm 0{\cdot}04)$ oersted.
contact U: $F = (0{\cdot}19 \pm 0{\cdot}01)$ oersted.

Errors are standard errors of the means.

These values are in good agreement notwithstanding the fact that each is less than 50% of the present field at the same latitude. It is unlikely that these low values arise as a result of the decay of the NRMs since the dyke and contact materials (basalt and red sandstone, respectively) are unlikely to possess the same decay characteristics.

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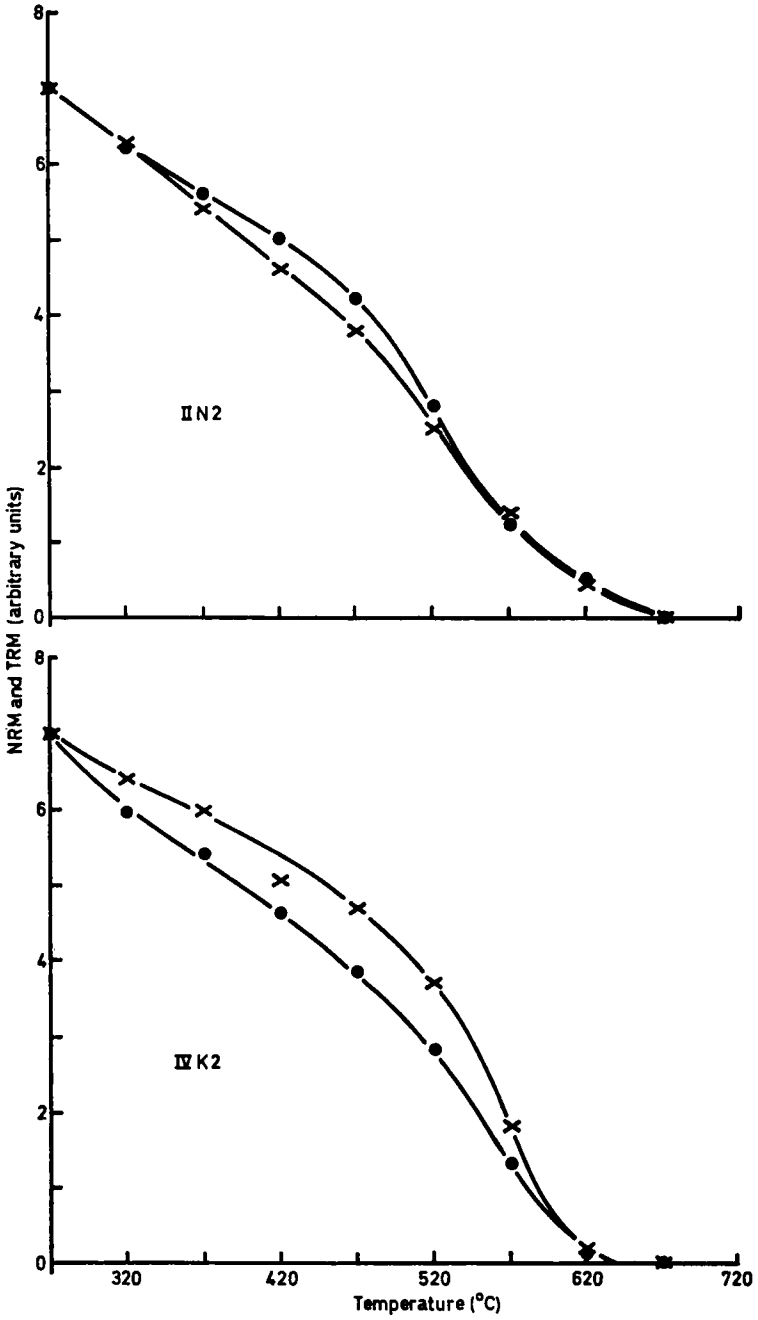


FIG. 4. Scottish baked contacts: examples of NRM and TRM curves normalized at 270 °C for each specimen.

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4.5. *Final intensity values for Scottish rocks*

Final intensity values and virtual dipole moments are shown in Table 2.

Table 2

Mean field intensities and virtual dipole moments for Scottish rocks. N=number of specimens, F=ancient field intensity, σ=estimate of standard deviation for N specimens, P=virtual dipole moment, All dyke–contact systems are reversed except N.

System	*N*	Mean *F* (oe)	σ (oe)	σ (%)	*P* ($\times 10^{-25}$ gauss.cm^3)
A	1	0·19	—	—	3·34
C	2	0·28	0·035	12·5	4·92
D	1	0·17	—	—	3·00
E	1	0·35	—	—	6·16
F	2	0·27	0·014	5·2	4·75
G	2	0·51	0·085	16·7	8·96
K	1	0·25	—	—	4·40
L	1	0·20	—	—	3·52
M	1	0·27	—	—	4·75
N	2	0·25	0·030	12·0	4·40
P	2	0·25	0·035	14·0	4·40
Q	2	0·64	0·042	6·6	11·30
S	2	0·52	0·021	4·0	9·15
U	3[1]	0·20	0·023	11·5	3·52
V	2	0·45	0·042	9·3	7·92
W	1[2]	0·19	—	—	3·34
Y	2	0·027	0·008	10·1	1·39
Z	1	0·056	—	—	0·99

[1] Includes one dyke specimen; [2] dyke specimen

From each of ten of the baked contacts it was possible to measure two specimens and hence the quoted intensity value is the mean of two values (except for system U, which includes a dyke specimen and hence the intensity is the mean of three values) The virtual dipole moments were calculated using the paleomagnetic inclination (±59·3°) obtained from the same suite of rocks (Smith 1966a). The mean VDM is $4{\cdot}94 \times 10^{25}$ gauss.cm^3 with a standard deviation of $\pm 2{\cdot}58 \times 10^{25}$ gauss.cm^3 (±52·3%) and a standard error of the mean of $\pm 0{\cdot}61 \times 10^{25}$ gauss.cm^3 (±12·4%).

The mean of the estimated standard deviations in intensity for all twenty-nine specimens (twenty-seven contacts, two dykes) at the three temperatures to which equation (1) was applied was ±10·2% (Table 1). The mean of the estimated standard deviations in intensity values from pairs of specimens from the ten contacts from which two specimens were measured (including the three specimens of system U) was ±10·6% (Table 2). The mean saturation magnetization change for contacts and dykes was −(13·0±9·0)%, where the ±9·0% error is the standard deviation which contributes to the spread of results. Further, the dyke and contact intensity values from system U lie within ±7·5% of each other.

The above standard deviations do not arise from independent sources of error and hence it is not possible to calculate an accurate overall standard deviation in the VDMs due to alteration and experimental error. The figures indicate a maximum expected overall standard deviation of ±17·0% which contrasts with the overall standard deviation of the measured VDMs of ±52·3%. It will be shown later that this large standard deviation in the measured VDMs is unlikely to be caused by the effects of non-dipole fields and hence that it is most likely to be caused by fluctuations in the dipole strength.

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5. Ancient field intensities: Icelandic baked laterites

In view of the consistency of the Scottish field intensity determinations and the good agreement between the values from dyke U and its contacts, it was thought desirable to extend the work to a second Tertiary suite but from a different area at a different latitude and comprising different types of rock. Accordingly, baked laterites were sampled in western Iceland (specimen numbers prefixed with letter W) and eastern Iceland (no prefix). In this case the baking rocks were plateau basalt lavas. As before, samples were taken from within six centimetres of the contact faces.

Exactly the same measurements were made as before, both for the main intensity determinations and the supplementary tests. All heatings and coolings were carried out in air. Twenty-nine specimens from fifteen different contacts were initially utilized. Generally viscous components were larger than those in the Scottish contacts but T_s was always less than 270 °C. It was thus possible to apply equation (1) at 270 °C, 320 °C and 370 °C as before. An example of the vector heating diagrams from one of the laterites is shown in Fig. 5. NRMs were generally an order of magnitude higher than in the Scottish contacts. The saturation magnetization was high enough to enable complete heating and cooling curves to be plotted for each specimen.

Two baked laterite specimens were rejected because of obvious alteration during heating. In one case the high field Curie point changed by −40 °C, the NRM and TRM curves were of different shape and the zero field Curie point changed by −80 °C. In the other case the $J_{sat} - T$ curves indicated two magnetic constituents during heating, having Curie points of 500 °C and 600 °C, but only one constituent of Curie point 620 °C during cooling. The shapes of the NRM and TRM curves were also dissimilar. In each other case (twenty-seven specimens from thirteen contacts) the high field Curie point changed by less than ±10 °C (all Curie points were ⩾ 560 °C), the zero

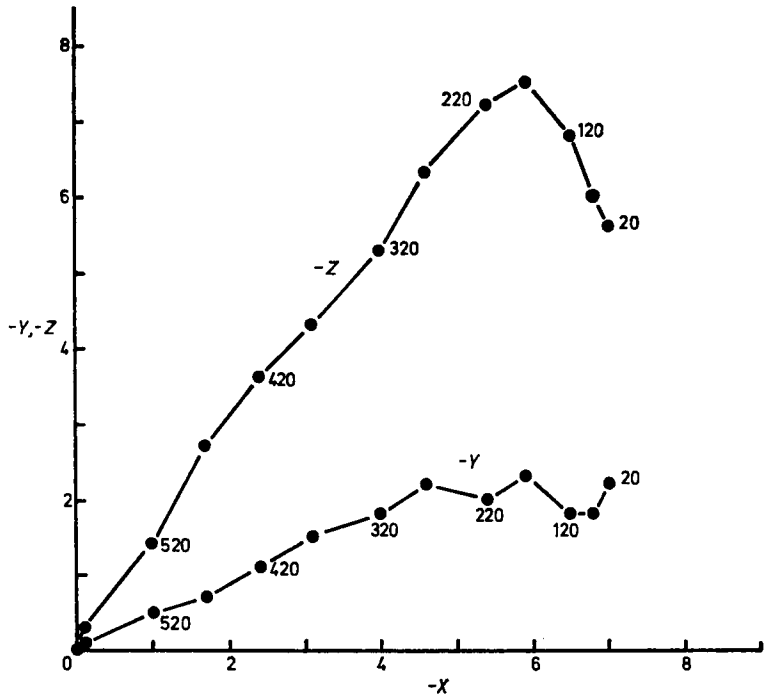


FIG. 5. Vector heating diagrams for Icelandic baked laterite (WB–6–3–1) in terms of three orthogonal components of magnetization (X, Y, Z in arbitrary units). Points are at 50 °C intervals: and numbers adjacent to points represent temperatures in °C.

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field Curie point changed by less than ± 40 °C, the NRM and TRM curves were similar in shape, and the heating and cooling J_{sat}–T curves were similar in shape and intensity.

For the twenty-seven accepted specimens the field intensity values are shown in Table 3 and the mean intensities and VDMs for each of the thirteen contacts are given in Table 4.

Table 3

Basic intensity data for the Icelandic baked laterites. F=ancient field intensity, σ=estimated standard deviation of F values at three temperatures.

No.	Specimen	F (oe) (370 °C)	F (oe) (320 °C)	F (oe) (270 °C)	Mean F (oe)	σ (oe)	σ (%)
1	WA8–3–1	0·23	0·24	0·25	0·24	0·010	4·2
	WA8–3–2	0·14	0·15	0·17	0·15	0·016	10·7
2	WA14–3–1	0·20	0·21	0·22	0·21	0·010	4·8
	WA14–3–2	0·22	0·23	0·25	0·23	0·016	7·0
3	WB3–2–1	0·75	0·70	0·69	0·71	0·032	4·5
	WB3–2–2	0·56	0·54	0·58	0·56	0·020	3·6
4	WB6–3–1	0·071	0·079	0·083	0·078	0·006	7·7
	WB6–3–2	0·066	0·069	0·072	0·069	0·002	2·9
	WB6–3–5	0·065	0·071	0·074	0·070	0·005	7·2
5	WC54–3–1	0·13	0·13	0·11	0·12	0·012	10·0
	WC54–4–1	0·16	0·13	0·12	0·14	0·021	15·0
6	WC64–3–1	0·24	0·26	0·28	0·26	0·020	7·7
	WC64–3–2	0·20	0·23	0·25	0·23	0·026	11·3
7	WC75–3–1	0·25	0·27	0·29	0·27	0·020	7·4
	WC75–3–2	0·28	0·30	0·32	0·30	0·020	6·7
	WC75–3–3	0·22	0·22	0·24	0·23	0·012	5·2
8	WC76–3–1	0·93	0·97	0·97	0·96	0·023	2·4
	WC76–3–2	0·73	0·79	0·82	0·78	0·045	5·8
9	WK17–3–1	0·42	0·46	0·45	0·44	0·021	4·8
	WK17–3–3	0·40	0·43	0·43	0·42	0·017	4·0
10	C9–3–1	0·31	0·31	0·30	0·31	0·007	2·3
	C9–3–2	0·36	0·36	0·36	0·36	0·000	0·0
11	C10–4–1	0·12	0·12	0·13	0·12	0·007	5·8
12	C11–3–1	0·35	0·32	0·34	0·34	0·016	4·7
	C11–4–1	0·35	0·32	0·34	0·34	0·016	4·7
13	C12–4–1	0·36	0·37	0·37	0·37	0·007	1·9
	C12–5–1	0·35	0·37	0·36	0·36	0·010	2·8

Table 4

Mean field intensities and virtual dipole moments for Icelandic baked laterites. N=number of specimens, F=ancient field intensity, σ=estimate of standard deviation for N specimens, P=VDM.

System	Polarity	N	Mean F (oe)	σ (oe)	σ (%)	P ($\times 10^{-25}$ gauss.cm^3)
WA8–3	N	2	0·20	0·064	32·0	3·12
WA14–3	R	2	0·22	0·014	6·4	3·44
WB3–2	N	2	0·64	0·106	16·5	10·00
WB6–3	R	3	0·072	0·005	6·9	1·13
WC54–3/4	N	2	0·13	0·014	10·8	2·03
WC64–3	N	2	0·25	0·021	8·4	3·90
WC75–3	R	3	0·27	0·032	11·9	4·22
WC76–3	R	2	0·87	0·100	11·5	13·60
WK17–3	R	2	0·43	0·014	3·3	6·71
C9–3	R	2	0·33	0·035	10·6	5·15
C10–4	R	1	0·12	—	—	1·87
C11–3/4	R	2	0·34	0·000	0·0	5·32
C12–4/5	R	2	0·37	0·007	1·9	5·78

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The reliability statistics were:

(i) Mean standard deviation in field intensity for twenty-seven specimens at 270°C, 320°C and 370°C = ±5·8%.

(ii) Mean J_{sat} change for thirteen specimens (one from each baked layer) = −(10·7±4·5)%. J_{sat} values before and after heating are plotted in Fig. 6.

(iii) Mean standard deviation in field intensity for pairs or triplets from twelve contacts (for one of the thirteen contacts one specimen only was measured) = ±10·6%. These figures indicate an overall maximum expected standard deviation of ±13·0% due to experimental error and alteration.

The mean VDM is $5{\cdot}12 \times 10^{25}$ gauss.cm^3 with a standard deviation of $\pm 3{\cdot}34 \times 10^{25}$ gauss.cm^3 (±65·4%) and a standard error of the mean of $\pm 0{\cdot}91 \times 10^{25}$ gauss.cm^3 (±17·8%). The overall standard deviation in the measured VDMs is thus again much higher than that estimated from the effects of alteration and experimental error. As for the Scottish case, this is unlikely to be caused by non-dipole field noise and probably represents fluctuations in the dipole strength (Section 8).

Virtual dipole moments were calculated using the mean palaeomagnetic inclination (±67·2°). The palaeodirectional data will be considered in greater detail in Section 7.

6. Ancient field intensities: Icelandic lavas

Tertiary field intensities have also been obtained from plateau basalt lavas collected from western and eastern Iceland as part of the complete palaeomagnetic survey of

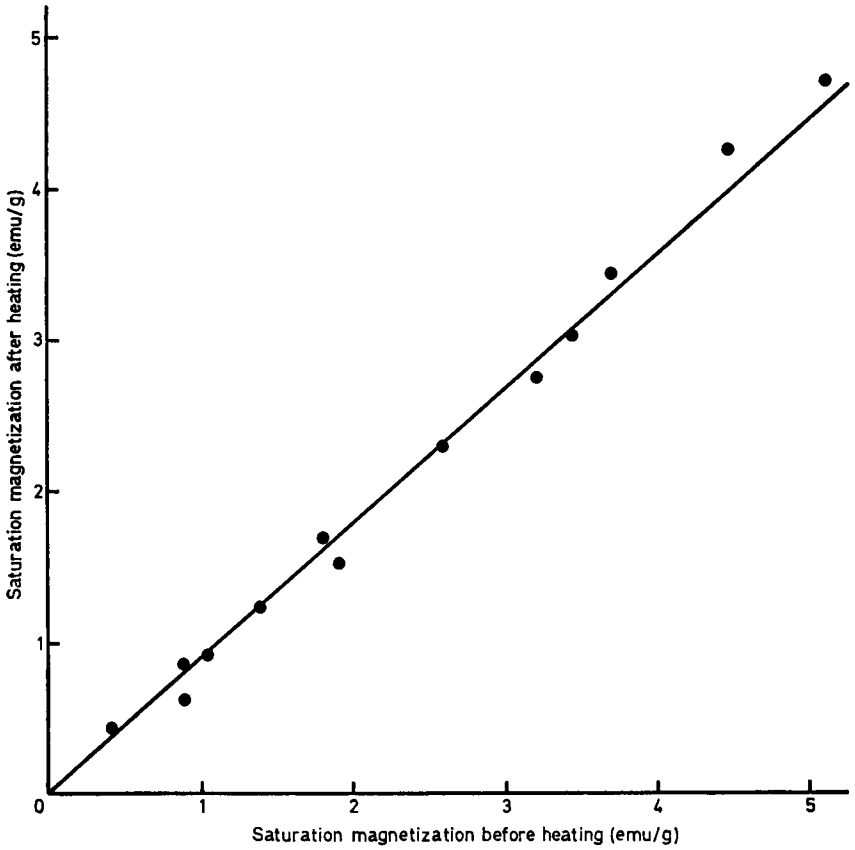


FIG. 6. Icelandic baked laterites: saturation magnetization before heating plotted against that after heating to above the Curie point. J_{sat} in e.m.u./g. Line represents mean change in J_{sat} of −10·7%.

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Icelandic Tertiary activity at present being carried out in Liverpool, London and Iceland (Wilson *et al.* 1966). Wilson & Watkins (to be published) have classified basalt lavas into five types on the basis of a study of the opaque minerals present. Highly oxidized basalts in Class 5, in particular, are characterized by high magnetic stability when heated in air (Wilson & Haggerty 1966). Class 5 lavas exist among the plateau basalts from Iceland; and hence eighteen specimens (one from each of eighteen lavas) were selected for field intensity determination.

For these lavas a different technique, based on a.c. rather than thermal demagnetization, was used. The method was basically that developed by Van Zijl (1961). Each specimen was a.c. demagnetized at 50-oersted peak field intervals up to a maximum of 300 oersteds. It was then heated to 670 °C in air and cooled back to room temperature in a field of 0·60 oersted. The resulting TRM was then a.c. demagnetized at the same a.c. field intervals as for the NRM.

The ancient field intensity was deduced by applying equation (1) to the 100, 150, 200, and 250 oersted field values. NRM demagnetization curves indicated that all viscous components were removed below 100 oersteds. Intensity values and virtual dipole moments are given in Table 5.

Table 5

Intensity values and virtual dipole moments for Icelandic lavas. F=ancient field intensity, σ=estimate of standard deviation at four a.c. fields, P=VDM.

Rock	Polarity	F (oe) (~100)	F (oe) (~150)	F (oe) (~200)	F (oe) (~250)	Mean F (oe)	σ (oe)	σ (%)	P ($\times 10^{-25}$ gauss.cm^3)
WA4–1a	N	0·22	0·20	0·20	0·19	0·20	0·013	6·5	3·20
WA16–1a	N	0·41	0·41	0·42	0·43	0·42	0·010	2·4	6·72
WB3–2a	N	0·36	0·38	0·37	—	0·37	0·010	2·7	5·93
WB10–2a	N	0·34	0·35	0·37	0·40	0·37	0·026	7·0	5·93
WC54–2a	N	0·22	0·22	0·22	—	0·22	0·000	0·0	3·52
WC75–1a	R	0·30	0·30	0·29	0·29	0·30	0·008	2·7	4·81
WG10–1a	N	0·41	0·41	0·43	0·46	0·43	0·024	5·6	6·89
WG11–1a	R	0·42	0·44	0·44	0·48	0·45	0·025	5·6	7·21
WG12–2a	R	0·39	0·40	0·40	0·42	0·40	0·013	3·2	6·40
WG13–2a	R	0·37	0·37	0·37	0·38	0·37	0·006	1·6	5·93
WK4–1a	*	0·10	0·11	0·11	0·12	0·11	0·008	7·3	1·76
WL3–1a	*	0·32	0·35	0·36	0·40	0·36	0·033	9·2	5·76
F50–1a	N	0·43	0·43	0·44	0·44	0·44	0·008	1·8	7·05
F55–2a	N	0·47	0·49	0·52	0·56	0·51	00·39	7·6	8·18
K31–Fla	R	0·30	0·34	0·34	0·36	0·34	0·025	7·3	5·60
K35–1a	R	0·25	0·25	0·24	0·25	0·25	0·006	2·4	4·00
M30–1b	N	0·20	0·20	0·21	0·21	0·21	0·008	3·8	3·36
N7A–1a	N	0·44	0·43	0·44	0·46	0·44	0·013	3·0	7·05

* The directions of magnetization of these two specimens are south (declination) and down (inclination). It is possible that they were magnetized during reversals of polarity.

Reliability tests were carried out as before, except that zero field Curie points were not available. Heating and cooling saturation magnetization curves were similar in shape, and no high field Curie point changed within ±10 °C on heating in air. NRM and TRM curves were also similar in shape. All high field Curie points were ⩾ 500 °C. The reliability statistics were:

(i) Mean standard deviation in intensity at four fields (100, 150, 200, 250 oersteds) for all eighteen specimens = ±4·5%.

(ii) Mean J_{sat} change during heating in air for all eighteen specimens = −(6·4±4·3)%. J_{sat} values before and after heating are plotted in Fig. 7.

These values indicate a maximum standard deviation of ±7% in the final result due to alteration and experimental error.

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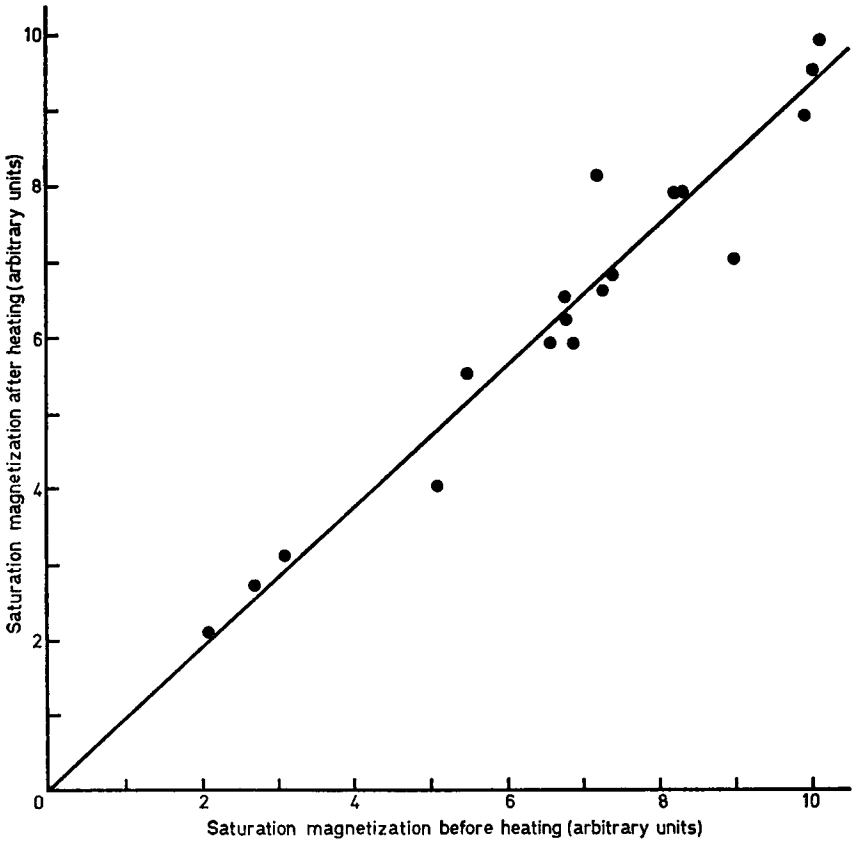


FIG. 7. Icelandic Class 5 lavas: saturation magnetization before heating plotted against that after heating to above the Curie point. J_{sat} in arbitrary units. Line represents mean change in J_{sat} of $-6{\cdot}4\%$.

The mean virtual dipole moment is $5{\cdot}50 \times 10^{25}$ gauss.cm^3 with a standard deviation of $\pm 2{\cdot}12 \times 10^{25}$ gauss.cm^3 ($\pm 38{\cdot}6\%$) and a standard error of the mean of $\pm 0{\cdot}50 \times 10^{25}$ gauss.cm^3 ($\pm 9{\cdot}1\%$). VDMs were calculated using the mean palaeomagnetic inclination ($\pm 64{\cdot}7\%$) (see Section 7). Again the overall standard deviation of the measured VDMs is much greater than that estimated from the effects of alteration and experimental error. As before this is likely to be due to fluctuation in the dipole strength rather than non-dipole field noise.

For all the Icelandic rocks (eighteen lavas and thirteen baked laterites) the mean VDM is $5{\cdot}34 \times 10^{25}$ gauss.cm^3 with a standard deviation of $\pm 2{\cdot}52 \times 10^{25}$ gauss.cm^3 ($\pm 47{\cdot}2\%$) and a standard error of the mean of $\pm 0{\cdot}45 \times 10^{25}$ gauss.cm^3 ($\pm 8{\cdot}4\%$).

7. Summary of palaeodirections and ages of Icelandic specimens

It was not possible to orient all of the baked laterite samples in the field owing to the crumbly nature of some, and limited accessibility. Fourteen were oriented, and for these, directions of magnetization were deduced from the high temperature regions of the vector heating diagrams. The tops were marked on four further specimens so that the polarities could be deduced from the signs of the inclinations. Nine specimens were completely unoriented, but came from beneath lavas of known orientation. Directions were also obtained from fifteen a.c. cleaned specimens of

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lava baking the laterites. The mean inclination (±67·2°) of all of the above twenty-nine laterite and lava specimens was used to calculate the VDMs of the baked laterites.

All of the Class 5 lava specimens were oriented. Directions were calculated after demagnetization to 150 oersteds (minimum scatter field). The mean inclination (±64·7°) of the eighteen lava directions was used to calculate the VDMs of the lavas. All directional data are summarized in Table 6. The northern hemisphere virtual geomagnetic pole position deduced from all forty-seven specimens is 74·0°N latitude, 0° longitude.

Table 6

Summary of Icelandic palaeodirectional data. N=number of specimens, D=mean declination of remanent magnetization (normal), I=mean inclination of remanent magnetization, R=length of the resultant of the N unit vectors, k=estimate of Fisher's precision parameter, α_{95}=semiangle of cone of 95% confidence for mean direction, δ=angular standard deviation of mean direction,. All forty-seven specimens give a mean virtual geomagnetic pole position (northern hemisphere) of λ=74°N, φ=0°.

Specimens	*N*	*D* (deg.ETN)	*I* (± deg.)	*R*	*k*	α_{95} (deg.)	δ (deg.)
Baked laterites	14	5·0	66·7	13·47	24·3	8·2	15·9
Associated lavas	15	33·6	66·3	10·71	3·3	25·2	44·4
Baked laterites+ associated lavas	29	17·8	67·2	24·06	5·7	12·4	33·9
Class 5 lavas	18	353·9	64·7	16·12	9·0	12·2	26·4
All specimens	47	7·7	66·7	40·04	6·6	8·8	31·6

Dr R. L. Grasty of Cambridge University has kindly dated three of the western Icelandic lavas using the potassium–argon method. Details are given in Table 7.

Table 7

Age data for Icelandic lavas. Obtained by Dr R. L. Grasty of Cambridge University.

Rock	K_2O%	Volume of radiogenic A^{40} (mm^3/g)	Atmospheric contamination	Age (m.y.)
WA3–3	0·44	$5·62\times10^{-5}$	86·9%	3·8±1·6
WA3–3	0·44	$3·02\times10^{-5}$	91·8%	2·1±0·7
WK15–1	0·37	$5·90\times10^{-5}$	87·2%	4·8±0·6
WK15–1	0·37	$5·28\times10^{-5}$	83·1%	4·3±0·3
WC76–2	0·32	$3·65\times10^{-5}$	84·6%	3·4±1·0
WC76–2	0·32	$<3·68\times10^{-5}$	?	<4·5

The ages lay in the range (2·1 m.y.–4·8 m.y.). In connection with the Icelandic palaeomagnetic project mentioned previously, eight specimens, two from each of four lavas, have also been dated by Dr R. Grasty. The ages of these specimens lay in the range (11·0 m.y.–18·1 m.y.). Two further specimens, one from each of two different lavas, had ages of 21·9 m.y. and 22·0 m.y. but with errors greater than ±54%.

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8. Discussion and conclusions

The purpose of this paper is mainly to present the Tertiary field strengths and virtual dipole moments and to describe the methods and precautions taken in the field intensity determination. It is not proposed to discuss in detail the significance of the results at this stage, since this is best done by considering these and all previous determinations together. A complete analysis and review of all published field intensity data is at present being prepared for publication. Nevertheless, several points may be usefully discussed briefly here.

(a) The tertiary mean virtual dipole moments obtained here and those calculated from data in the literature are summarized in Table 8.

Table 8

Summary of Tertiary mean virtual dipole moments obtained in the present work and calculated from data in the literature. N=number of individual VDMs contributing, σ=standard deviation, ϵ=standard error of the mean.

Reference	Rocks	N	*VDM* ($\times 10^{-25}$ gauss.cm^3)	σ ($\times 10^{-25}$ gauss.cm^3)	σ (%)	ϵ ($\times 10^{-25}$ gauss.cm^3)	ϵ (%)
This work	Scottish baked contacts/dykes	17	4·94	2·58	52·3	0·61	12·4
This work	Icelandic baked laterites	13	5·12	3·34	65·4	0·91	17·8
This work	Icelandic lavas	18	5·50	2·12	38·6	0·50	9·1
This work	Icelandic combined	31	5·34	2·52	47·2	0·45	8·4
Momose 1963	Japanese tuffs, basalts and andesites	37	5·77	2·68	46·4	0·44	7·6
Nagata 1943 (approximate)	Japanese basalts and andesites	11	5·50	1·70	30·9	0·49	8·9
Koenigsberger 1938 (approximate)	Various igneous, mainly German	18	5·02	2·31	46·0	0·54	10·7
Johnson *et al.* 1948 (approximate)	Pacific ocean bottom sediments	8	5·28	0·71	13·4	0·25	4·7

All six of the mean VDMs lie within $\pm 8{\cdot}7\%$ of their mean value ($5{\cdot}31 \times 10^{25}$ gauss.cm^3). All three of the accurate mean VDMs (Icelandic combined and Scottish obtained here, and Japanese obtained by Momose) lie within $\pm 7{\cdot}9\%$ of their mean value ($5{\cdot}35 \times 10^{25}$ gauss.cm^3). These mean values may be contrasted with the present geomagnetic dipole moment which is $8{\cdot}0 \times 10^{25}$ gauss.cm^3. The results suggest that during the Tertiary the Earth's mean dipole moment was 33% weaker than it is at the present moment.

(b) The close agreement of the mean VDMs gives confidence in their validity. The three accurate mean VDMs were obtained from eight different rock types from three different areas of the world. Although the mean of these accurate determinations is 33% lower than the Earth's present dipole moment, it is unlikely that any distorting influence such as magnetic decay could have had such a uniform effect on so many rock types as to reduce mean VDMs to such close equality.

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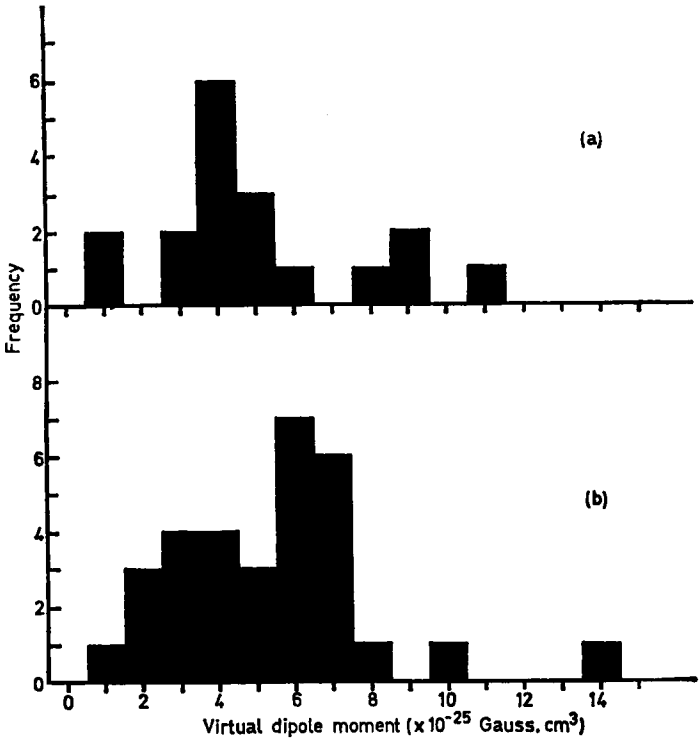


FIG. 8. Histograms of virtual dipole moments deduced from (a) Scottish baked contacts and dykes, and (b) Icelandic baked laterites and Class 5 lavas.

(c) The VDMs obtained here are plotted as histograms in Fig. 8 (Scottish and Icelandic separately) and in Fig. 9 (Scottish and Icelandic together and also separated into normal and reversed sets). Both the Scottish and Icelandic histograms are peaked in the region of the mean VDMs and possess large spreads giving rise to the large standard deviations listed in Table 8 (Scottish $\sigma = 52{\cdot}3\%$, Icelandic combined $\sigma = 47{\cdot}2\%$). The histograms of the normal and reversed sets are somewhat dissimilar, but the number of normal specimens is perhaps too small to give this dissimilarity definite significance.

It was shown previously that the large standard deviations of Tertiary VDMs were at least three times greater than would have been expected from the effects of alteration during heating and of experimental error. The question arises as to whether or not the large spreads merely represent non-dipole field noise. The 1945 intensity field has been analysed (Smith 1966b) whence it has been shown that if the non-dipole field strength has always been approximately the same proportion of the dipole field strength [as suggested by Creer (1962) and Irving *et al.* (1964) on the basis of directional analyses of the 1945 field and palaeomagnetic data] then the standard deviations of the VDMs are still a factor of two too high to be explained by non-dipole field noise. It is suggested here that the large spreads of VDMs arise because of actual variations of the dipole moment with periods at least an order of magnitude shorter than the period of reversal of the dipole ($\sim 10^6$ years in the recent past). This possibility will be examined in greater detail in the paper mentioned at the beginning of this section.

(d) It seems that certain classes of highly oxidized lavas and rocks may be heated in air without much alteration due to the heating, in order to determine ancient field

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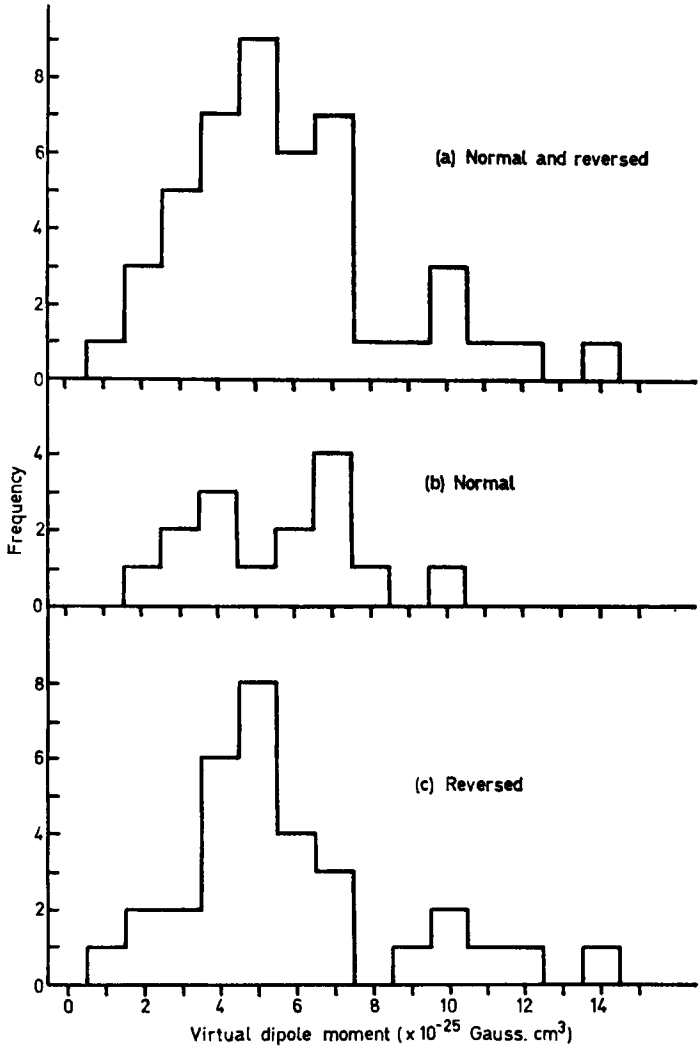


FIG. 9. Histograms of virtual dipole moments deduced from Scottish and Icelandic rocks: (a) normal and reversed together, (b) normal only, and (c) reversed only. Scottish VDMs have been normalized to the Icelandic VDMs through the mean values.

intensities. The technical problem now remaining is to discover how to heat less chemically and magnetically stable rocks without altering them and without changing the nature and amount of magnetic material present. One of the keys to the problem is the use of a good high-powered microscope for observing opaque magnetic minerals.

9. Acknowledgments

I would like to thank Dr R. L. Wilson for his advice during this work, Dr P. Dagley for writing the computer program used for the directional data, and the Science Research Council for a Research Assistantship during the tenure of which part of this

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research was carried out. The results presented here have been incorporated into a thesis submitted to the University of London.

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1966 *June.*

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